

Swiss Institute of Banking and Finance



University of St.Gallen

7. Capital Structure

Prof. Dr. Markus Schmid

3,130/4,130 Corporate Finance

Course Outline

1. General Information
2. Risk and Return
3. Portfolio Theory and CAPM
4. Corporate Valuation
5. Cost of Capital
6. Corporate Financing and the Issuance of Securities
- 7. Capital Structure**
8. Risk Management
9. Conflicts of Interest and Corporate Governance
10. Market Efficiency and Behavioral (Corporate) Finance
11. Empirical Research in Corporate Finance

After this lecture you will have understood...

- the mechanisms underlying **capital structure** (choices),
- why capital structure is **irrelevant** in a Modigliani and Miller world,
- what the **real world limitations** to the MM propositions are,
- how **taxes** and **bankruptcy costs** affect capital structure choice, and
- what **signal** is conveyed to the market by announcing different types of financing (e.g., internal financing, debt and equity issuances).

In the public press

THE WALL STREET JOURNAL.

English Edition | January 16, 2020 | Print Edition | Video

Home World U.S. Politics Economy Business Tech **Markets** Opinion Life & Arts Real Estate WSJ. Magazine

Subscribe | Sign In

€1 for 2 Months

Search 

Heavy Debt Crushed Owners of Toys ‘R’ Us

Interest payments of \$6 billion limited investment firms’ ability to improve stores, compete online

By [Miriam Gottfried](#) and [Lillian Rizzo](#)

Sept. 19, 2017 7:17 pm ET

 PRINT  TEXT

61 

Competition may have brought down Toys “R” Us Inc. But the debt that three Wall Street firms heaped on the company when they took it private—and the way the ownership group was constructed—left the retailer without a fighting chance.

When a consortium comprised of Vornado Realty Trust and private-equity firms KKR & Co. and Bain Capital bought the toy retailer for \$6.6 billion in 2005, it was already



Business Markets World Politics TV More

How \$5 billion of debt caught up with Toys ‘R’ Us

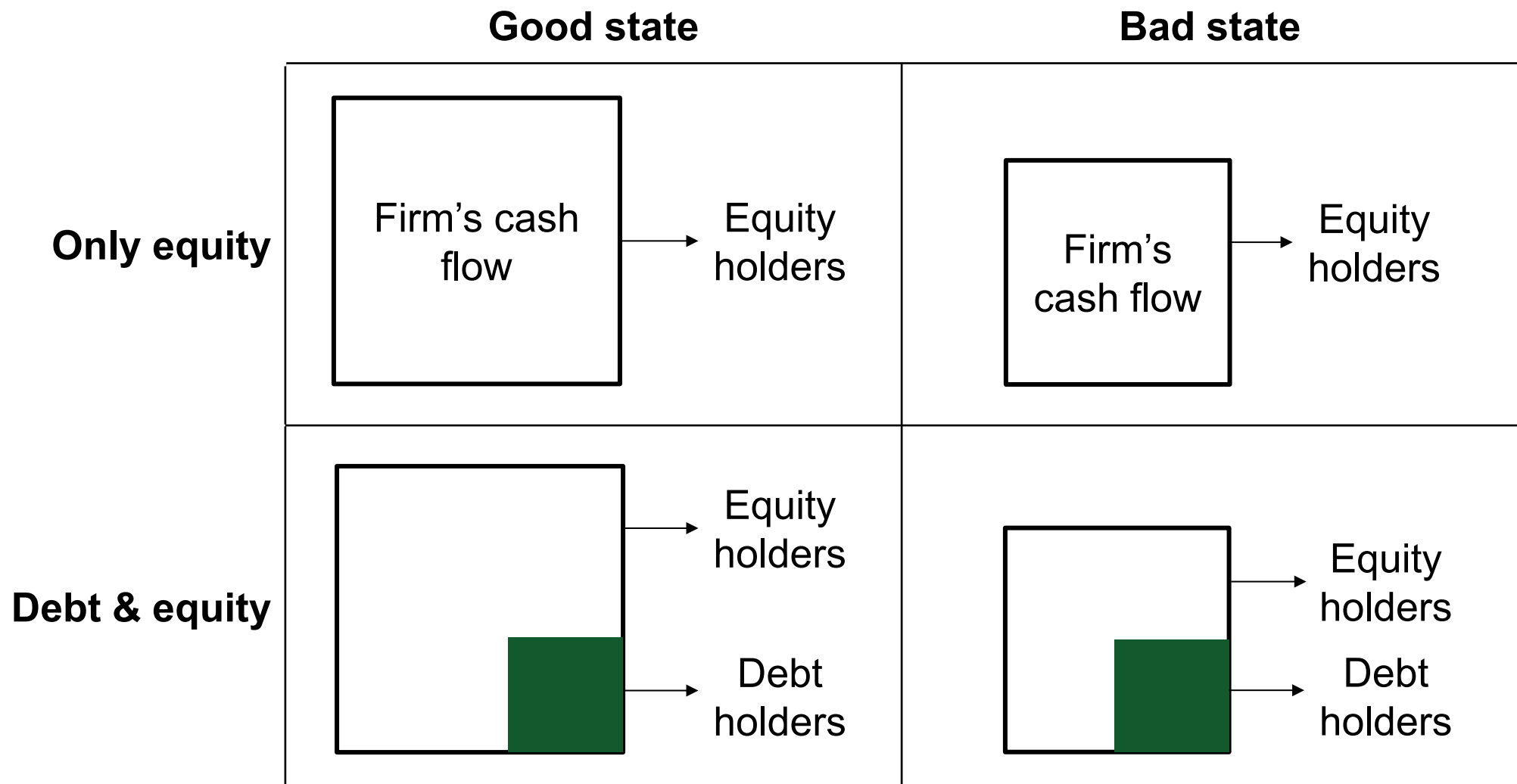
Jessica DiNapoli, Tracy Rucinski

5 MIN READ



(Reuters) - Toys “R” Us Inc has been making \$400 million in interest payments on its debt every year, largely due to its \$6.6 billion leveraged buyout in 2005. This week, it succumbed to its debt burden, leading to the biggest bankruptcy of a U.S. retailer since that of Kmart in 2004.

Introduction: Why is debt safer than equity?



From a “perfect” MM world to a world with frictions...

Modigliani & Miller

Assumptions about perfect markets and symmetric information, no taxes as well as no costs of financial distress.

taxes

Trade-off theory (part 1)

Taxes are introduced in the Modigliani & Miller world.

cost of financial distress

Trade-off theory (part 2)

Costs of financial distress are taken into account in the Modigliani & Miller world with taxes.

Pecking order theory

Asymmetric information (and **signaling**) influences the choice between internal, debt, and equity financing.

Assumptions of the Modigliani & Miller World

- **Perfect markets:** No transaction costs, perfect information, etc.
 - **Law of one price:** Assets with the same payoff have the same price.
 - Investors can borrow and lend at the **same rates as the firm**, and can thereby do (or «undo») a firm's financing decisions.
- **No taxes**
- **Payout** policy (dividends, repurchases) is ignored.
- Capital structure choice does **not affect** investment and operating policies.

Modigliani and Miller

No taxes (I)

Imagine two firms, one unlevered (U) and one levered (L). Both have the same stream of operating income. Imagine that you acquired 1% in both the unlevered and the levered firm, equity only in U and debt and equity in L.

Unlevered firm (U)		
Dollar Investment		Dollar Return
Debt	0	0
Equity	$0.01 \times E_U$	$0.01 \times \text{Profits}$
Total	$0.01 \times E_U$	$0.01 \times \text{Profits}$
	$= 0.01 \times V_U$	

Levered firm (L)		
Dollar Investment		Dollar Return
Debt	$0.01 \times D_L$	$0.01 \times \text{Interest}$
Equity	$0.01 \times E_L$	$0.01 \times (\text{Profits} - \text{Interest})$
Total	$0.01 \times (D_L + E_L)$	$0.01 \times \text{Profits}$
	$= 0.01 \times V_L$	

Source: BMAE (2022, p. 461)

⇒ Same Payoff! Therefore V_U must equal V_L .

Modigliani and Miller

No taxes (II)

If you are “risk-loving”, you could buy 1% of the equity in the levered firm.
As an alternative strategy you could also borrow $0.01 D_L$ on your own account and purchase 1% in the unlevered firm.

Levered firm (L)		
Dollar Investment		Dollar Return
Borrowing	0	0
Equity	$0.01 \times E_L$	$0.01 \times (\text{Profits} - \text{Interest})$
Total	$0.01 \times (V_L - D_L)$	$0.01 \times (\text{Profits} - \text{Interest})$

Financed investment in unlevered firm (L)		
Dollar Investment		Dollar Return
Borrowing	$-0.01 \times D_L$	$-0.01 \times \text{Interest}$
Equity	$0.01 \times V_U$	$0.01 \times \text{Profits}$
Total	$0.01 \times (V_U - D_L)$	$0.01 \times (\text{Profits} - \text{Interest})$

Source: BMAE (2022, p. 462)

⇒ Same Payoff! Therefore V_L must equal V_U .

Modigliani and Miller

Proposition I

- Investors can “**undo**” a firm’s financing policy.
- If the value of two assets is the same when combining them (value additivity), this also holds when splitting them.
- The value is determined by the **PVs of cash flows** generated by a firm’s assets, not by the securities issued to buy the assets.
- Absent taxes and bankruptcy risk, the weighted average cost of capital does not depend on the ratio between debt and equity.

MM Proposition I: With perfect capital markets and no taxes the value of the firm is independent of its capital structure.

Modigliani and Miller

Example for no taxes (I)

Example

- Company X reviews its capital structure. It has no debt, pays no taxes, and has access to perfect capital markets. The company has 100 shares outstanding, trading at \$20, thus a market value of \$2,000. It has no investment opportunities, hence the operating income is fully paid out to stockholders as dividends. The firm expects an operating income of \$250.
- The company's president comes to the conclusion that shareholders would be better off if the company had equal proportions of debt and equity. She proposes to issue \$1,000 of debt at the risk-free rate of 10% and use the proceeds to buy back 50 shares.

Modigliani and Miller

Example for no taxes (II)

Unlevered firm

Balance sheet

Assets	Equity
2000	2000

Income statement

Operating income	250	Profit	250
------------------	-----	--------	-----

Levered firm

Balance sheet

Assets	Debt
	1000
2000	Equity
	1000

Income statement

Operating income	250	Profit	150
Interest	-100		

Modigliani and Miller

Example for no taxes (III)

- The expected return on assets is the expected operating income as a percentage of the market value of all securities:

$$r_A = \frac{\text{expected operating income}}{\text{market value of all securities}}$$

- For the unlevered firm X, this is:

$$r_E = r_A = \frac{250}{2,000} = 0.125$$

- For the levered firm X with equal amounts of debt and equity, it is:

$$r_A = \frac{E}{E + D} r_E + \frac{D}{E + D} r_D$$

Modigliani and Miller

Example for no taxes (IV)

- Rearranging yields:

$$r_E = r_A + \frac{D}{E}(r_A - r_D)$$

- and an equally debt / equity financed firm's return of:

$$r_E = 0.125 + \frac{1,000}{1,000} \cdot (0.125 - 0.10) = 0.15$$

Modigliani and Miller

Example for no taxes (V)

	All Equity	Equal Debt and Equity
Price per share	\$20	\$20
Expected EPS $\left(\frac{\text{Operating income}}{\# \text{ shares}} \right)$	\$2.50	\$3.00
Expected ROE $\left(\frac{\text{EPS}}{\text{share price}} \right)$	12.5% $\longrightarrow$ 15%	ROE increases!

- Are equity investors better off when the firm is levered?
- **No**, because increases in expected return are exactly offset by an increase in risk and required rate of return (see subsequent slides).

Modigliani and Miller

Example for no taxes (VI)

- In reality, earnings are **uncertain**. Assume company X faces three possible scenarios. If the firm was all-equity financed, EPS and ROE would be as follows:

Market conditions	Bad	Neutral (expected outcome)	Good
Operating income	100	250	300
EPS $\left(\frac{\text{Operating income}}{\text{\# shares}} \right)$	1	2.5	3
ROE $\left(\frac{\text{EPS}}{\text{share price}} \right)$	5%	12.5%	15%

Modigliani and Miller

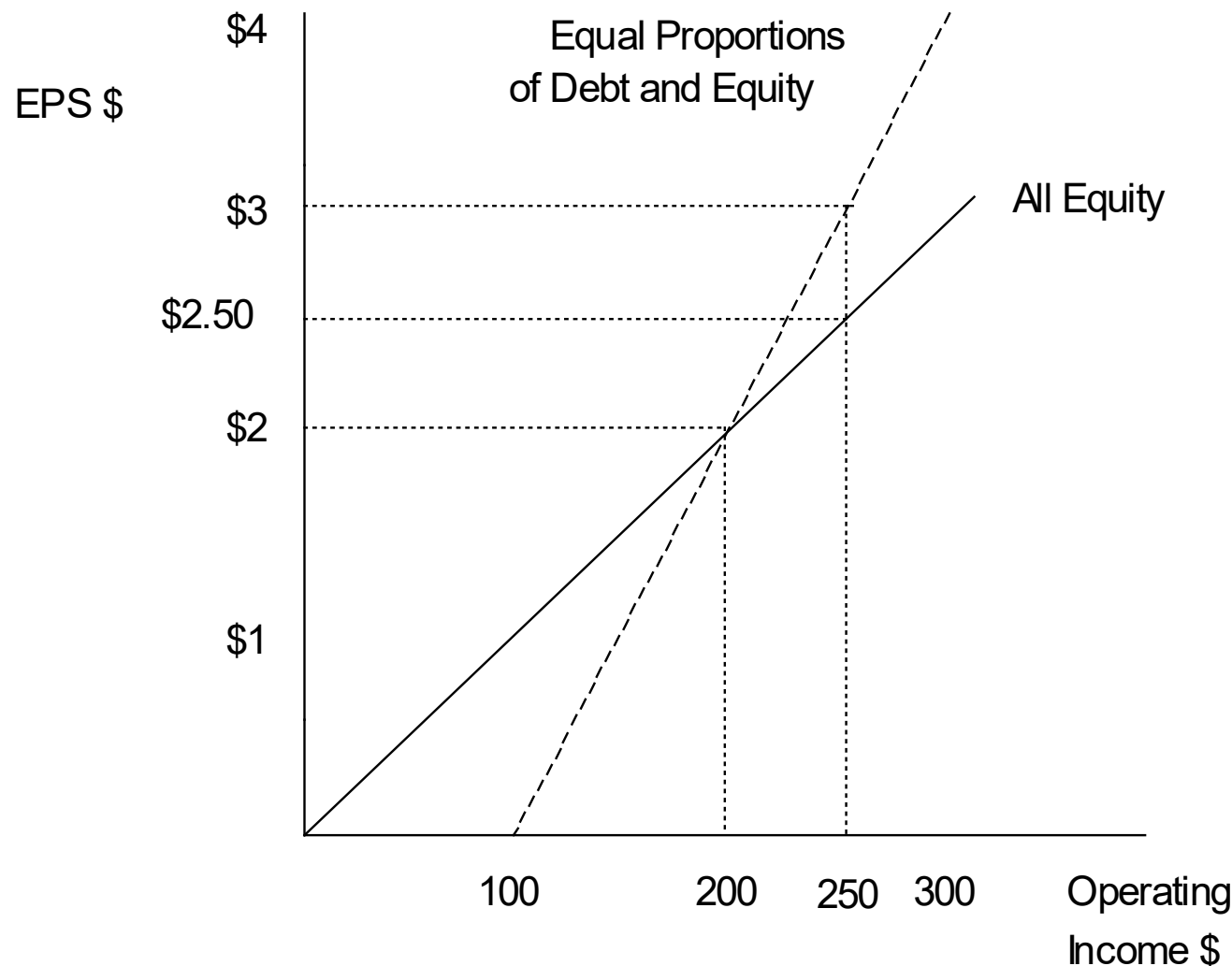
Example for no taxes (VII)

- If the firm was financed with **equal proportions** of equity and debt, this would result in the following returns:

Market conditions	Bad	Neutral (expected outcome)	Good
Operating income	100	250	300
Interest	100	100	100
EPS	0	3	4
Return on Equity (%)	0	15	20
Return on Debt (%)	10	10	10

Modigliani and Miller

Example for no taxes (VIII)



The two lines cross at the point where the return on the market value of assets is equal to the interest rate paid on debt.

Modigliani and Miller

Proposition II

- The effect of leverage depends on the company's earnings.

MM Proposition II: The expected return on equity of a levered firm increases with the D/E ratio. The rate of increase depends on the spread between the expected return on assets and the return on debt.

Modigliani and Miller

Proposition I vs. II

- These findings seem contradictory. While MM Proposition I states that the value of the company is independent of its capital structure, Proposition II states that increasing leverage increases the expected return on equity. What is right?
- **Both:** The fact that the expected return on equity increases with leverage does not mean that shareholders are better off. Rather, the expected (or “required”) return on equity **must** increase to compensate shareholders for the increase in risk brought about by the increase in leverage. This premium is equal to $(r_A - r_D)D/E$:

$$r_E = \text{return on assets} + \text{risk premium} = r_A + \boxed{\left(r_A - r_D\right) \frac{D}{E}}$$

- Any increase in expected return is exactly offset by an increase in risk and the required rate of return.

Modigliani and Miller

Equity risk (no taxes)

- Why does equity become riskier as leverage increases? Consider the difference in the expected return on X's equity between the “good” and “bad” outcomes:

	Bad	Good	Variation
Operating income	100	300	
<u>Unlevered firm</u>			
EPS	1	3	
ROE (%)	5	15	10%-points
<u>Levered firm</u>			
EPS	0	4	
ROE (%)	0	20	20%-points

Modigliani and Miller

Debt and equity risk (no taxes)

- The change in financial structure does not affect the amount or risk of the cash flows on the total **package** of debt and equity.
- Increasing the amount of debt also increases **debt holder risk**. Thus, the debt holders are likely to ask for a higher return.
- As the firm borrows more, more of the business risk is **transferred** from stockholders to bondholders.
- More borrowing also increases equity risk (see last slide) and the required return on equity increases as well.
- The weighted average return on debt and equity is, however, **unaffected**.

Modigliani and Miller

Debt and equity risk (no taxes) (II)

How does WACC change when D/E changes?

- Starting point: 30% debt, $r_E = 10\%$, $r_D = 5\%$

$$r_A = \frac{E}{E+D}r_E + \frac{D}{E+D}r_D = \frac{0.7}{0.7+0.3}0.1 + \frac{0.3}{0.7+0.3}0.05 = \underline{0.085}$$

- Now the company increases debt to 50%. What happens to r_E , assuming that r_D increases to 6%?

$$r_E = r_A + \frac{D}{E}(r_A - r_D) = 0.085 + 1(0.085 - 0.06) = \underline{0.11}$$

- What happens to the WACC?

$$r_A = \frac{0.5}{0.5+0.5}0.11 + \frac{0.5}{0.5+0.5}0.06 = \underline{0.085}$$

Modigliani and Miller

Summary

- Capital structure is irrelevant in a world with **perfect** capital markets and no taxes. Firm value is determined by a firm's real assets and not by the type of securities issued. If a firm increases the fraction of debt, equity becomes riskier.
- To compensate shareholders for the increase in risk, the firm must offer them a higher rate of return. This increase in the firm's cost of equity capital **offsets** any potential benefit from substituting “cheaper” debt for “more expensive” equity, leaving the firm's weighted average cost of capital and its market value unchanged.
- MM's propositions are a starting point. However, what is important in reality in determining optimal capital structure are market **imperfections** and taxes!

Modigliani and Miller

Limitations

- We observe various market imperfections:
 - Firms incur **costs for issuing** securities.
 - Firms and investors do **not** face identical borrowing and lending costs.
 - Pooling single investors' interests **reduces transaction costs**.
 - Corporate bond schemes allow to **circumvent regulatory constraints**.
 - The set of securities available to investors is **incomplete**.
 - Credit risk can lead to **conflicts of interest** between debt investors, equity investors, and management.

The trade-off theory

Taxes (I)

- The trade-off theory **relaxes the assumptions** of no taxes and no costs of financial distress.
- There is a **third** claimant to a firm's cash flows in addition to debt and equity holders: The government taxes cash flows to equity and debt differently.
 - Historically, interest paid on a firm's borrowing is seen as a business expense and is deductible from taxable income.
 - Dividends are treated as a return to the firm's owners and are therefore not tax deductible.
- This difference creates a bias towards debt. Holding before-tax cash flows fixed, debt finance reduces a firm's taxable income, resulting in higher after-tax cash flows.

The trade-off theory

Taxes (II)

- We can estimate the present value of this tax shield as follows:

$$\text{Interest payment} = \text{return on debt} \cdot \text{amount borrowed} = r_D \times D$$

$$\text{PV}(\text{tax shield})^* = \frac{\text{corporate tax rate} \cdot \text{interest payment}}{\text{expected return on debt}} = \frac{T_c r_D D}{r_D} = T_c D$$

* Assumption that the debt and interest will be constant until infinity.

- The cost of capital can be re-written to reflect this tax shield:

$$\text{After-tax WACC} = \frac{E}{E + D} r_E + \frac{D}{E + D} r_D (1 - T_c)$$

The trade-off theory

Taxes – Example (I)

Example

Firms U and L are nearly identical. Both have a corporate tax rate of 30% and an annual operating income of \$100, forever with certainty. Firm U has no debt while firm L has issued \$200 in debt at $r_f=10\%$.

	Firm U	Firm L
Operating income	100	100
Interest	0	20
Pre-tax income	100	80
Taxes	30	24
Net income (goes to shareholders)	70	56
MV equity	700	560
MV debt	0	200
MV firm	700	760

The trade-off theory

Taxes – Example (II)

Firm L's annual tax bill is \$6 less than firm U's. These tax savings constitute a tax shield (like the one from depreciation). We can discount the annual tax shield at the **same rate** as that of the **interest payments** generating it – in this case we use the risk-free rate – yielding a PV of $6/0.1 = 60$. This is the difference in market values between firms U and L. We thus have:

$$\text{Value of levered firm (V}_L\text{)} = \text{Value of unlevered firm (V}_U\text{)} + \text{PV tax shield (T}_c\text{D)}$$

What does this imply for value maximization?

The trade-off theory

Costs of financial distress

- Debt ratios in reality are rather low, between 30% and 40% on average, so there must be costs to debt financing. The trade-off theory proposes bankruptcy costs, and more generally, costs of financial distress as the main costs of debt finance.
- In the MM world, when the value of equity falls to zero, debt holders take over the firm. There should be no costs to bankruptcy because there is no reduction in the cash flows generated by the company.
- In practice, there may be sizeable costs of bankruptcy. These costs make debt less attractive because they reduce the expected value of the firm's cash flows.

The trade-off theory

Costs of financial distress – Real world example

- **Chapter 11** is a chapter of Title 11, the U.S. Bankruptcy Code, which permits reorganization under the bankruptcy laws of the U.S.:
 - When a business is unable to service its debt or pay its creditors, the business or its creditors can file with a federal bankruptcy court for protection under Chapter 11.
 - In general, the debtor remains in control of its business operations as a *debtor in possession*, and is subject to the oversight and jurisdiction of the court.
 - A Chapter 11 bankruptcy will result in one of three outcomes for the debtor, reorganization, conversion to Chapter 7 bankruptcy, or dismissal.
 - In order for a chapter 11 debtor to reorganize, they must file (and the court must confirm) a plan of reorganization. Such a plan is a compromise between the major stakeholders in the case, including, but not limited to the debtor and its creditors
- **Example:** «*Toys ‘R’ Us wins confirmation of chapter 11 payment plan*», The Wall Street Journal (November 13, 2018).



The trade-off theory

Costs of financial distress – Real world example (II)

- Toys "R" Us Inc. began as a small, local toy store in Washington, D.C., in 1948. At its height, it operated or licensed about 2,000 stores in 38 countries.
- Toys "R" Us filed for bankruptcy protection in September 2017, hoping to trim its chain of stores and survive; this filing was supposed to get Toys "R" Us on solid financial ground and diminish the debt left over from its \$6.6bn LBO acquisition in 2005 by a consortium led by private equity firms: Bain Capital, KKR, and Vornado Realty Trust.
- Toys's 10-K for 2017 reports \$4.6bn in long-term debt out of \$6.9bn in total assets, resulting in a long-term debt ratio of: $4.6\text{bn} / 6.9\text{bn} \approx 67\%$; \$385m interest payments were paid during the fiscal year that ended in January 2017.
- However, the turnaround effort failed leaving 33,000 workers without jobs and all 800 U.S. stores closed. The firm abandoned the effort to reorganize and failed to emerge from bankruptcy. A group of five hedge funds that owned a critical piece of secured debt with a face value of \$668m pushed Toys into liquidation in March 2018 after a disastrous holiday season.

The trade-off theory

Costs of financial distress – Real world example (III)

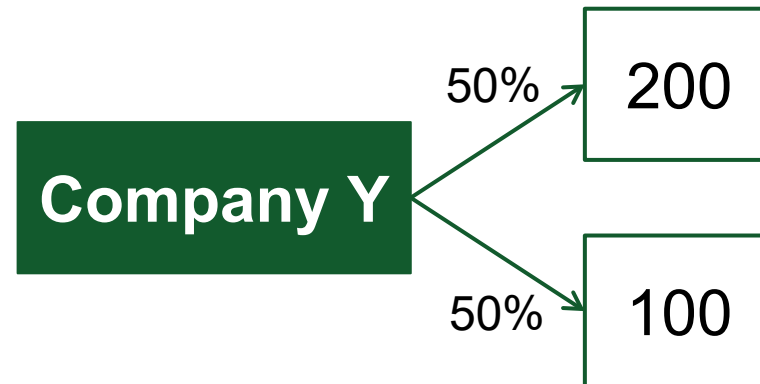
- The group of hedge funds that controlled Toys "R" Us's secured debt deny to be responsible and attribute failure to market forces, such as the boom in online sales harmful for bricks-and-mortar retailers.
 - Cash from the sales of inventory, businesses in Asia and some real estate helped pay off top tiers of debt; suppliers and other junior creditors, that owed about \$800 million, agreed to a settlement that will give them an estimated 22 cents for each dollar they are owed.
 - Disagreements continue over the intellectual property controlled by the hedge funds which is still needed by continuing businesses in Asia that are operated by joint ventures; Toys Asia employs 11,000 people.
- ⇒ High debt levels are risky: While a fully equity-financed firm can easily absorb a “bad holiday season”, a heavily indebted firm may become unable to service its debt obligations and end up under Chapter 11.

The trade-off theory

Costs of financial distress – Example (I)

All-equity case

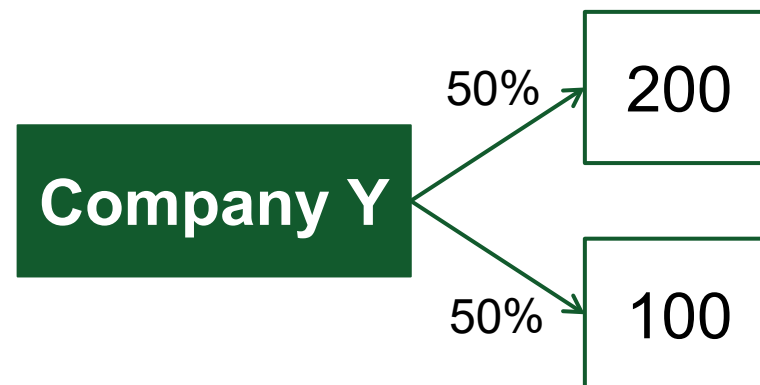
Company Y will generate either 100 or 200 (with equal probability) every year starting next year. The company has no systematic risk, the risk-free rate is 10%.



$$E = \frac{200 \cdot 0.5 + 100 \cdot 0.5}{0.1} = 1,500$$

Levered case

The leveraged company has debt with a face value of \$1,500. The company goes into default if CF = 100. Does this change the value of the company relative to if it was all equity financed?



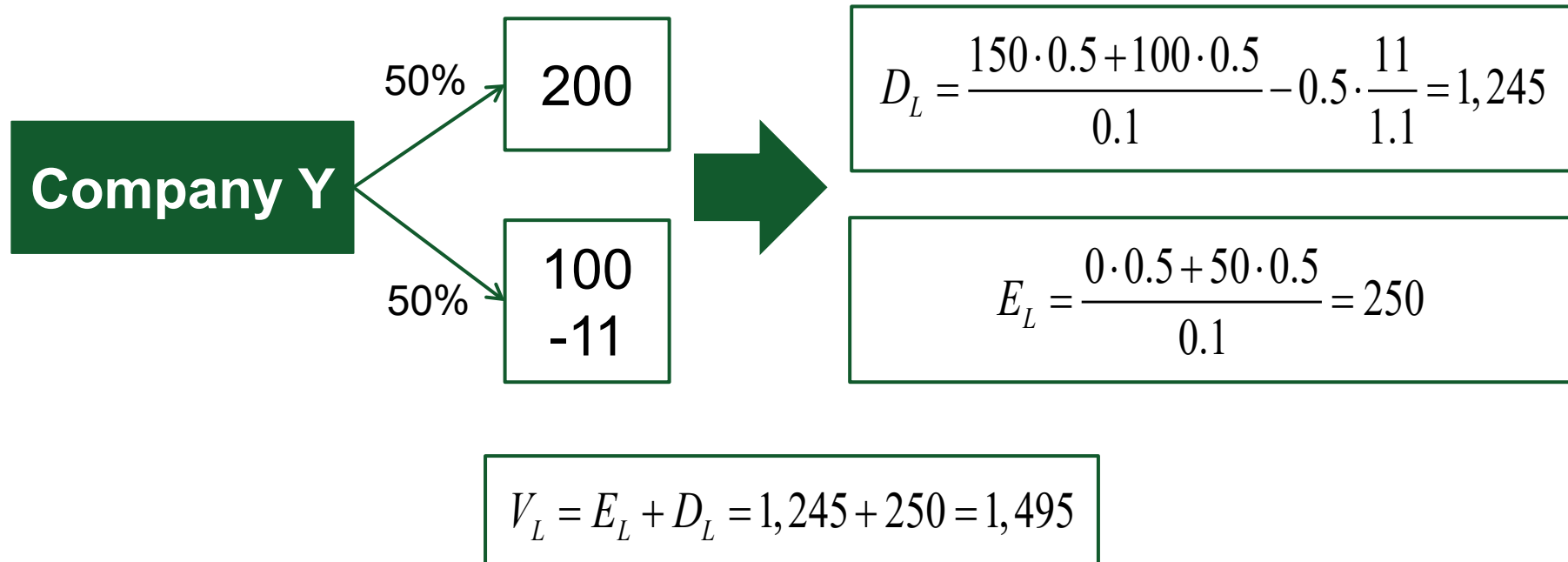
$$D_L = \frac{150 \cdot 0.5 + 100 \cdot 0.5}{0.1} = 1,250$$

$$E_L = \frac{50 \cdot 0.5 + 0 \cdot 0.5}{0.1} = 250$$

The trade-off theory

Costs of financial distress – Example (II)

Suppose there is a one-time \$11 bankruptcy cost in one year. That means that if the company fails to meet its debt service obligation of \$150, it will have to pay a fee of \$11 (e.g., lawyer's fees)



Firm value has decreased by the PV of the bankruptcy costs!

The trade-off theory

Direct and indirect costs of bankruptcy

Academic research suggests that these costs are relatively small and too small to account for the low observed debt ratios.

Direct costs

- Lawyers and bankruptcy judges are examples of direct costs of financial distress.

Indirect costs

- Managerial attention **diverted** from managing assets to managing liabilities.

- Firm loses **flexibility** when monitored closely by creditors.
- Assets must be sold in **fire-sales**. Firms with highly specific assets obtain the worst prices. Especially true if assets do not have value outside the industry and the industry is cyclical (oil).
- **Intangible** assets may be destroyed if the firm is broken up or sold. Firms with high advertising or R&D expenditure have lower than average debt ratios (tech firms). Firms with high expenditure on property, plant & equipment have higher than average debt ratios (hotels).

The trade-off theory

Summary

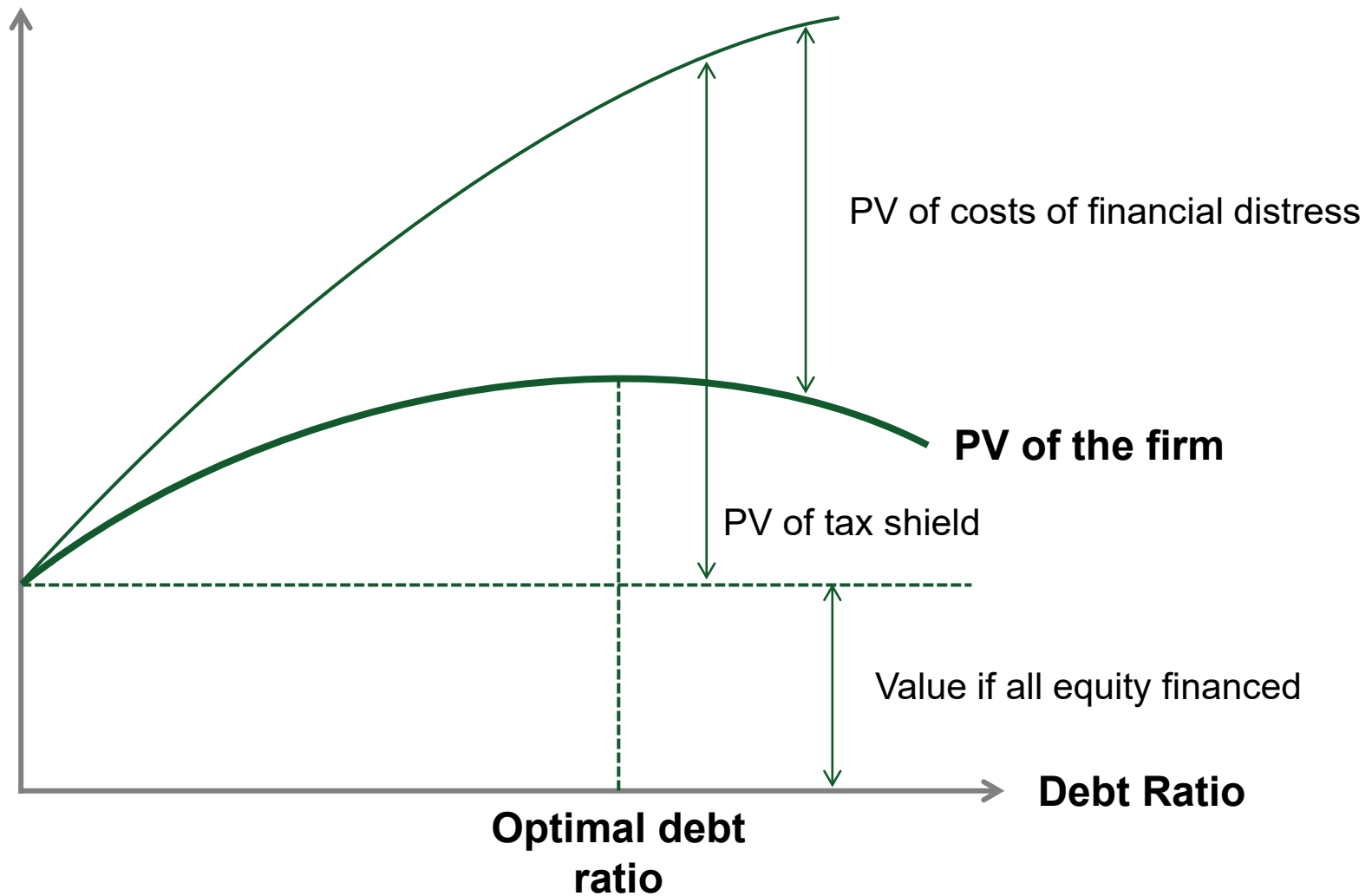
- According to the trade-off theory, debt finance has both benefits (tax shield) and costs (expected costs of financial distress):

$$\begin{aligned}\text{Value of levered firm (V}_L\text{)} &= \text{Value of unlevered firm (V}_U\text{)} \\ &\quad + \text{PV tax shield (T}_c\text{D)} \\ &\quad - \text{PV expected costs of fin. distress}\end{aligned}$$

- Starting out from a situation where the firm is unlevered, a small increase in leverage has benefits (tax shield) but virtually no costs as the probability of financial distress remains negligible.
- As leverage increases, the probability of financial distress becomes more important and eventually so large that firm value decreases if leverage is increased any further. At this point, the firm has reached its **optimal capital structure**.

The trade-off theory

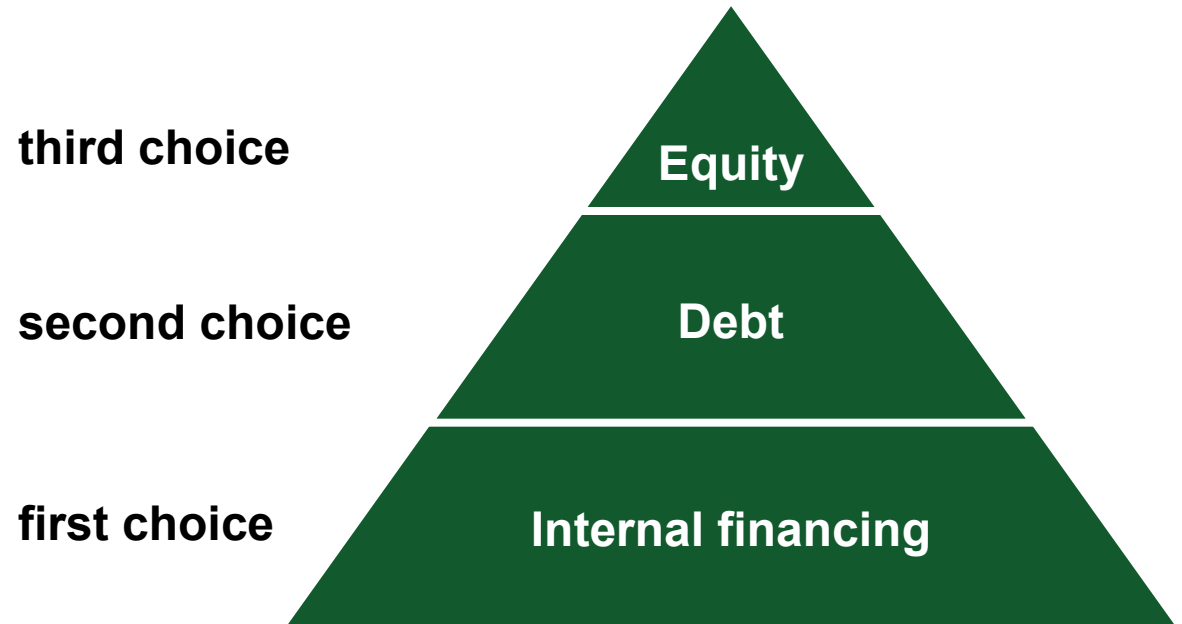
Graphical interpretation



The pecking order theory

The fact that managers know more about their companies' prospects, risks, and value than outsiders results in **asymmetric information**.

Asymmetric information affects the **choice** between internal and external financing and between new issues of debt and equity securities. Asymmetric information problems lead to a pecking order, in which an investment is financed:



New equity issues are a **last resort** when the company runs out of debt capacity, because of costs of financial distress.

The pecking order theory

Example (I)

- To the outside world, Tom & Co. and Jerry Inc. are identical. Both firms are successful and have good growth opportunities. However, both firms are risky and expectations are frequently bettered or disappointed. Current expectations price shares at \$100.

	Tom & Co.	Jerry Inc.
True value could be higher	\$120	\$120
Best current estimate	\$100	\$100
True value could be lower	\$80	\$80

- If both firms need to raise capital, they can choose between issuing new debt and issuing new common stock.

The pecking order theory

Example (II)

Issue of debt

Tom's managers believe that its shares currently trade **below** the actual value (underpriced).



The firm would prefer to issue debt, not underpriced equity!

Issue of \$120m 5-yr senior notes

Issue of shares

Jerry's managers believe that its shares currently trade **above** the actual value (overpriced).



The firm would prefer to issue equity, not debt that has to be serviced!

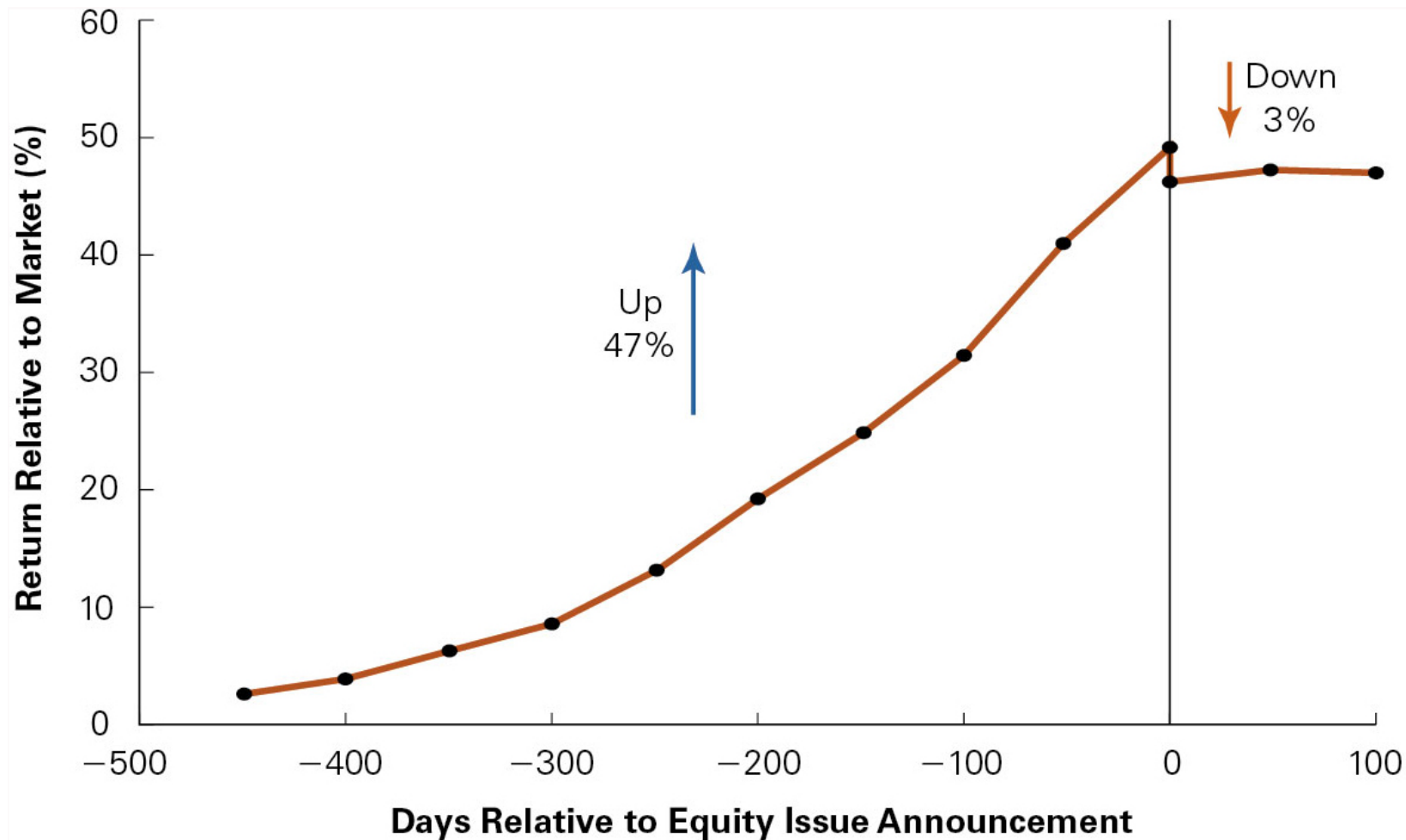
Issue of 1.2m new shares of common stock (i.e., \$120m)

The decision to issue stock alone implies that it must be overvalued so that the market will discount the share price upon announcement.

As a result, both managers will favor debt issues over equity issues.

What, do you think, is the market reaction to equity issues in reality?

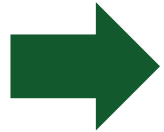
Raising equity capital in a SEO



Source: Berk & DeMarzo (2020), Figure 16.3, p. 620

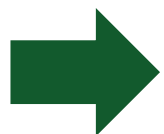
The pecking order theory

The pecking order and its implications



If external finance is required, firms issue the **safest** security first. That is, they start with debt, then possibly hybrid securities, and finally equity as a last resort.

- No well-defined target debt-equity mix exists.
- The observed debt ratios reflect each firm's cumulative requirement for external financing.
- The most profitable firms borrow less, because they are less dependent on outside financing. The attraction of interest tax shields is of secondary importance.
- Debt ratios change when there is an imbalance of internal cash flow, net of dividends, and real investment opportunities.



The pecking order theory explains the **inverse** (intra-industry) relationship between **profitability** and financial **leverage**.



The trade-off and pecking order theory

Some evidence

- Rajan and Zingales (1995) found that debt ratios depend on four main factors:

Factor	Observation	Trade-off theory	Pecking order theory
Size	Larger firms tend to have higher debt ratios	Large firms have a lower risk of financial distress	
Tangible assets	Firms with high ratios of fixed assets to total assets have higher debt ratios	Firms with more tangible assets have a lower costs of financial distress	
Profitability	More profitable firms have lower debt ratios		Profitable firms rely on internal finance
Market to book	Firms with higher ratios of market-to-book value have lower debt ratios	Growth opportunities increase risk of financial distress	Higher MTB implies profitability, thus higher reliance on internal finance

More recent evidence on leverage determinants

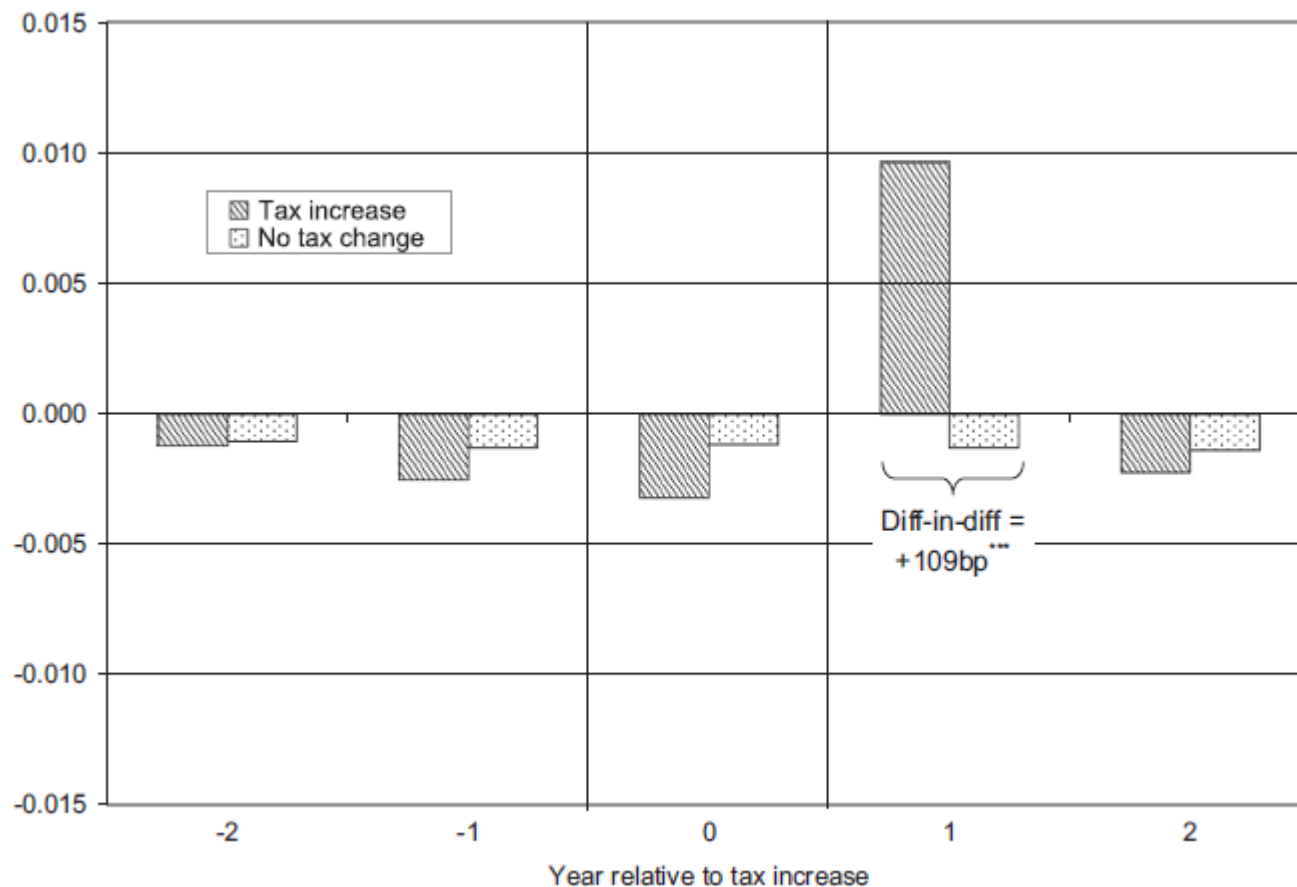
Heider and Ljungqvist (2015)

- The problem of most research: Debt / leverage is endogenously determined. Results represent correlations, no causality.
- Heider and Ljungqvist (2015) propose a novel empirical approach that exploits state-wise changes in corporate income tax rates.
- They use state-level changes in corporate income taxes (affecting different companies at different points in time) to investigate the effect of taxes on leverage.
- Example: To control for industry- and state-specific developments, they compare leverage changes among North Carolina firms in 1991 (tax raise) to contemporaneous changes in leverage among firms that operate in the same industry but are located in states without tax changes in 1991, say in South Carolina.

More recent evidence on leverage determinants (II)

Heider and Ljungqvist (2015)

- They show that taxes have a first-order effect on capital structure: On average, firms increase leverage by around 40 basis points for every percentage-point tax increase.



Key takeaways

1. Modigliani and Miller's propositions assert that financial leverage has **no** effect on shareholder wealth: Shareholders' expected return increases as the firm's debt-equity ratio increases, to compensate for the extra equity risk.
2. In reality, we observe various market **imperfections**. Considering costs of bankruptcy, conflicts of interest, information problems, and incentive effects, debt policy **does** matter.
3. According to the trade-off theory, the manager chooses a capital structure by weighting **benefits** (tax shield) and **costs** (expected costs of financial distress) of debt finance.
4. The **pecking order theory** accounts for the signaling effect (resulting from asymmetric information).

Mandatory readings

- Brealey, R., Myers, S., Allen, F., and A. Edmans, 2022, Principles of Corporate Finance, 14th Edition, New York: McGraw-Hill, Chapter 16 (Does Debt Policy Matter?), Chapter 17 (How Much Should a Corporation Borrow?), & 18 (Financing and Valuation). Note: Chapter 18 also covers material from classes 4-6.

Additional readings

- Heider, F., and A. Ljungqvist, 2015, As certain as debt and taxes: Estimating the tax sensitivity of debt from state tax changes, *Journal of Financial Economics* 118, 684-712.
- Graham, J.R., and M.T. Leary, 2011, A review of empirical capital structure research and directions for the future, *Annual Review of Financial Economics* 3, 309-345.
- Rajan, R.G., and L. Zingales, 1995, What do we know about capital structure? Some evidence from international data, *Journal of Finance* 50, 1421-1460.